

Power BI Migration Core for Nexus

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Problem Framing

LEGACY SYSTEM

- ▶ Oracle Warehouse Builder (ETL)
- ▶ SAP BusinessObjects (Reporting)
- ▶ 20+ years of accumulated complexity

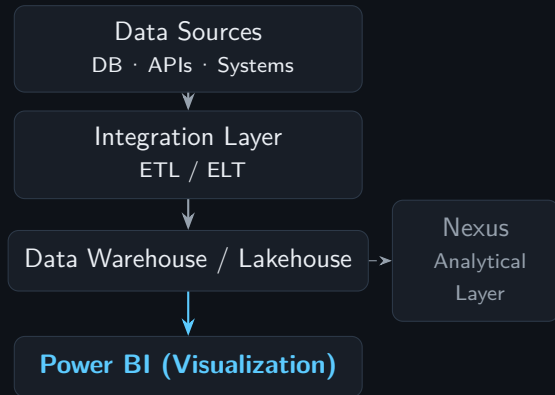
CHALLENGES

- ▶ Fragmented ETL and reporting layers
- ▶ Limited scalability and flexibility
- ▶ No path to analytics or AI integration

Replace with a modern, scalable, unified data platform

Target Architecture

ARCHITECTURE LAYERS



KEY PRINCIPLE

- ▶ Power BI = **Visualization Layer only**
- ▶ Logic and transformations **live upstream**
- ▶ Clear separation: Ingestion · Storage · Viz

Migration Strategy

PHASE 1 — INVENTORY

- ▶ Catalogue reports and ETL jobs
- ▶ Classify: Critical / Replaceable / Obsolete
- ▶ Identify data owners

PHASE 2 — PARALLEL BUILD

- ▶ New system runs alongside legacy
- ▶ Define clear integration points
- ▶ Validate via pilot reports
- ▶ **No Big Bang**

PHASE 3 — CONTROLLED CUTOVER

- ▶ Stepwise migration (per report cluster)
- ▶ Rollback always available
- ▶ Legacy stays active (transition phase)

Risks & Mitigation

TECHNICAL

- ▶ Hidden ETL dependencies
- ▶ Data inconsistencies
- ▶ KPI definition mismatch
- ▶ Performance deviations

ORGANIZATIONAL

- ▶ User resistance (20+ years habits)
- ▶ Misaligned expectations
- ▶ Training gap

MITIGATION

- ▶ Automated validation (old vs. new)
- ▶ Early key-user involvement
- ▶ Versioned datasets
- ▶ Remove unused reports

Role of Power BI

STRENGTHS

- ▶ Fast dashboard development
- ▶ Interactive analytics
- ▶ Microsoft ecosystem integration

TECHNICAL

- ▶ Import Mode (in-memory)
- ▶ DirectQuery (live data)
- ▶ DAX (calculation layer)

Power BI = Visualization Layer

The Calyr / Nexus Analogy

THE PARALLEL

- ▶ Calyr: scientific data platform built on Nexus
- ▶ Administrative data platform built on Power BI
- ▶ **Same principle** — typed objects, declared contracts, governed outputs

NEXUS CONTROL LOOP

- ▶ Declare → Build → Test → Score → Publish
- ▶ Experiments are **declared**, not improvised
- ▶ Outputs are valid only if traceable

Data Warehouse = Nexus for administrative data

Nexus Core Structure

CLASSICAL BI

Tables → Dashboards One
domain · one tool · one
query Efficient, but limited

NEXUS APPROACH

Data → Typed Objects →
Models

- ▶ Time domain (kinetics, SPR)
- ▶ Reciprocal space (SAXS)
- ▶ Real space (geometry)
- ▶ Administrative domain (KPIs, reports)

CORE STRUCTURE

- ▶ Contracts (schemas + typed interfaces)
- ▶ Validation (declared checks, traceability)
- ▶ Orchestration (declare → build → test → score → publish)

Reports declared, not
improvised → Analytical
Platform

Final Vision

FROM

- ▶ Static reporting
- ▶ Manual ETL processes
- ▶ Isolated data silos

TO

- ▶ Scalable data platform
- ▶ AI-ready infrastructure
- ▶ Integrated analytics
- ▶ Nexus foundation

Data Warehouse = Foundation